

BitLocker in Azure Local



Agenda

- **Objective:** Explore BitLocker functionality and management in Azure Local.
Scope: The end-to-end encryption lifecycle, key protector creation & storage, Portal vs. PowerShell management, and break-glass recovery.



BitLocker Fundamentals

- **Purpose:**
 - Protect against **data theft/exposure** on **lost/stolen** or **poorly decommissioned** systems.
- **TPM-backed security:**
 - Best protection with TPM 1.2+
 - Enables **pre-boot integrity** checks.
- **Non-TPM options:**
 - PIN/USB startup key or OS volume password
 - Weaker assurance than TPM

Management and protection scopes

- **Portal visibility and management:**
 - BitLocker settings are visible in Azure Portal once the **Microsoft Cloud Security Benchmark (MCSB)** initiative is applied.
 - Management is performed via PowerShell.
- **Two protection scopes:**
 - OS volumes (Boot)
 - Data volumes (CSV)

Azure Local and BitLocker Modules

- Traditional BitLocker tools and PowerShell cmdlets:
 - `manage-bde`, `Get-BitLockerVolume`, etc.
 - Not cluster-aware; require manual volume mapping.
- Azure Local module `AzureStackBitLockerAgent` adds cluster-aware cmdlets:
 - `Get-ASBitLocker` (state)
 - `Enable-ASBitLocker` (encrypt)
 - `Disable-ASBitLocker` (decrypt)
- **Scopes:**
 - `-Local`, `-Cluster`, `-PerNode`, `-MountPoint` targets specific CSV.

Traditional vs. Azure Local BitLocker Cmdlets

- **Traditional (`Get-BitLockerVolume` + scripts):**
 - Deep audits, raw protector metadata, custom reporting.
 - More suitable for forensics and edge cases
- **Azure Local (`Get/Enable/Disable-ASBitLocker`)**
 - Cluster-aware operations with minimal scripting.
 - Ideal for routine tasks

Get-ASBitLocker

- Querying Boot volumes and CSVs (local vs. per-node)
- BootVolume state (local):
 - `Get-ASBitLocker -VolumeType BootVolume -Local`
- CSV state (local):
 - `Get-ASBitLocker -VolumeType ClusterSharedVolume -Local`
- CSV state (per-node cluster-wide):
 - `Get-ASBitLocker -VolumeType ClusterSharedVolume -PerNode`
- CredSSP/RDP needed for `-PerNode` queries.

Enable-ASBitLocker

- Boot and CSV, Local vs. Cluster, and mount point targeting
 - Pre-condition: CSVs must be Online.
- Boot volumes (requires TPM 2.0)
 - `Enable-ASBitLocker -VolumeType BootVolume -Local`
- All CSVs on local node:
 - `Enable-ASBitLocker -VolumeType ClusterSharedVolume -Local`
- Specific CSV:
 - `Enable-ASBitLocker -VolumeType ClusterSharedVolume -Local -MountPoint "C:\ClusterStorage\Volume1"`
- All CSVs cluster-wide (CredSSP):
 - `Enable-ASBitLocker -VolumeType ClusterSharedVolume -Cluster`

Disable-ASBitLocker

- Mirror process of encryption with redirected I/O during background decryption
- Disable on BootVolume or CSV:
 - Same `-Local / -Cluster / -MountPoint` patterns.
- Lifecycle mirrors encryption:
 - Suspend → decrypt → resume, with redirected I/O until completion.

The CSV Encryption Lifecycle

- **1) Maintenance Mode (I/O paused; VMs paused):**
 - Cluster suspends the CSV resource to ensure consistency before changing encryption state.
- **2) Encryption Initiated (keys created):**
 - BitLocker starts and creates key protectors;
 - The volume remains suspended to avoid I/O during initial critical steps.
- **3) Resume (CSV online):**
 - Cluster brings the CSV back online, workloads resume while encryption continues.
- **4) Redirected I/O (performance impact):**
 - While encryption runs in background, non-owner nodes route all I/O through the coordinator (owner) node, potentially reducing performance.
- **5) Direct I/O (fully restored):**
 - When encryption completes, the CSV returns to normal I/O pathing, and performance normalization follows.

Phase 1: Maintenance Mode

- **Trigger:**
 - **Enable-ASBitLocker** (CSV) or traditional **Suspend-ClusterResource** places the CSV in maintenance.
- **Effect:**
 - I/O paused, VMs paused
 - Guarantees stability before key creation.
- **Verify:**
 - Cluster GUI or PowerShell shows **maintenance state**
 - Client access interrupted by design.

Phase 2: Encryption Initiated

- BitLocker begins and creates two protectors automatically:
 - Recovery Password (escrow into AD for domain-joined or Key Vault for non-domain)
 - External Key (.BEK) at `C:\Windows\Cluster` for auto-unlock during failover
- Volume state:
 - Still in maintenance while initial steps commit.

Phase 3: Resume (CSV Online)

- **Cluster resumes the CSV:**
 - VMs resume and I/O restarts.
- **Encryption continues in background**
 - CSV is accessible again.
- **Validation:**
 - Check resource comes online
 - Confirm workload access.

Phase 4: Redirected I/O (Performance Considerations)

- **Mechanism:**
 - Non-owner nodes **route all I/O** to the **owner (coordinator)** node during background encryption.
- **Impact:**
 - Higher latency possible
 - Depends on **workload** and **volume size**.
- **Mitigation:**
 - If needed, consider **temporarily relocating heavy workloads** to the **owner node** or throttling IO-intensive operations until completion.

Phase 5: Direct I/O (Back to Normal)

- **Signal:**
 - Background encryption **finishes**
 - CSV switches back to **Direct I/O**.
- **Post-checks:**
 - Verify **protector status**, **cluster events**, and **workload performance normalization**.
- **Outcome:**
 - **Fully restored** pathing;
 - Encryption state now **steady-state**.

Recovery Keys: Retrieval & Storage

- PowerShell (cluster-aware):
 - `Get-ASRecoveryKeyInfo | ft ComputerName, PasswordID, RecoveryKey`
- AD retrieval:
 - Use `Get-ADObject` to enumerate `msFVE-RecoveryInformation` entries.
- Recommendation:
 - Export and store keys in secure external locations (e.g., Azure Key Vault).

Non-AD CSV Protector (Cluster DB)

- **New functionality:**
 - Cluster can generate & maintain its own BitLocker protector in the local cluster database.
- **Recommendation:**
 - System (Boot) volumes should be BitLocker-protected;
 - Otherwise, Failover Cluster logs warning events (e.g., Event 1824).
- **Benefit:**
 - Works even when AD is unavailable or slow.

Provisioning: Internal vs External Keys

- **Internal recovery key:**
 - Protector stored in **cluster DB**; simpler; resume with normal flow.
- **External key file (.BEK):**
 - Create locally, **copy to shared path**, and **resume** with `-RecoveryKeyPath`
 - File can be removed afterward.
- **Tip:**
 - Use **Internal** by default
 - **External** for air-gapped or special compliance.

Failure Simulation & Recovery

- **Simulate loss:**
 - Remove CSV and cluster resource then re-add → expected **failure**.
- **Recover keys:**
 - Compare AD objects (**msFVE-RecoveryInformation**) with local protector to find missing key; or use **Get-ASRecoveryKeyInfo**. Configure **registry-based delegation** and run CredSSP authenticated remote commands to retrieve AD recovery info.
- **Unlock & online:**
 - **Start-ClusterPhysicalDiskResource -Name "Cluster Disk 1" -RecoveryPassword "<key>" -Cluster <ClusterName>**
 - **Resume-ClusterPhysicalDiskResource -Name "Cluster Disk 2" -RecoveryKeyPath \\Server\Share\F03E...BEK**
- **Rename & re-add CSV:**
 - Restore proper resource name and add back to CSV.

The logo for Opsgility features a black cloud icon on the left, which partially encloses the letters 'op'. The letters 's', 'g', 'i', 'l', 'i', 't', and 'y' are in a blue, lowercase, sans-serif font. The word 'Opsgility' is written in a lowercase, sans-serif font, with 'op' in black and 'sgility' in blue.

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